

Wrangling practice: one table

DATA 202 21FA, based on datasciencebox.org

Q&A

| When are `ggplot2` vs `dplyr` used in the real world?

Usually they're used *together*: wrangle-wrangle-wrangle-plot. Reports often include both *tables* and *graphics*.

| Are `mutate`, `select`, etc. universal?

These are only `tidyverse`, but SQL has very similar concepts.

| How to connect and join tables?

Next week! (You're welcome to read ahead.)

| Can you pipe too much?

No intrinsic limit, but: split into functions, simplify your analysis.

Today: Practice thinking about grammar of transformation

Friday: in-lab practice

Data: Hotel bookings

- Data from two hotels: one resort and one city hotel
- Observations: Each row represents a hotel booking
- You can try it: [Application Exercise](#)

```
hotels <- read_csv("data/hotels.csv")
head(hotels)
```

```
## # A tibble: 6 × 32
##   hotel          is_canceled lead_time arrival_date_ye... arrival_date_r
##   <chr>          <dbl>      <dbl>          <dbl> <chr>
## 1 Resort Hotel      0        342          2015 July
## 2 Resort Hotel      0        737          2015 July
## 3 Resort Hotel      0         7          2015 July
## 4 Resort Hotel      0        13          2015 July
## 5 Resort Hotel      0        14          2015 July
## 6 Resort Hotel      0        14          2015 July
## # ... with 27 more variables: arrival_date_week_number <dbl>,
## #   arrival_date_day_of_month <dbl>,
## #   stays_in_weekend_nights <dbl>, stays_in_week_nights <dbl>,
## #   adults <dbl>, children <dbl>, babies <dbl>, meal <chr>,
## #   country <chr>, market segment <chr>
```

Source: [TidyTuesday](#)

ascending / descending order

```
hotels %>%  
  select(adults, children, babies)  
  arrange(babies)
```

```
## # A tibble: 119,390 × 3  
##   adults children babies  
##   <dbl>     <dbl>   <dbl>  
## 1         2         0       0  
## 2         2         0       0  
## 3         1         0       0  
## 4         1         0       0  
## 5         2         0       0  
## 6         2         0       0  
## # ... with 119,384 more rows
```

ascending / descending order

```
hotels %>%  
  select(adults, children, babies)  
  arrange(babies)
```

```
## # A tibble: 119,390 × 3  
##   adults children babies  
##   <dbl>     <dbl>  <dbl>  
## 1         2         0      0  
## 2         2         0      0  
## 3         1         0      0  
## 4         1         0      0  
## 5         2         0      0  
## 6         2         0      0  
## # ... with 119,384 more rows
```

```
hotels %>%  
  select(adults, children, babies)  
  arrange(desc(babies))
```

```
## # A tibble: 119,390 × 3  
##   adults children babies  
##   <dbl>     <dbl>  <dbl>  
## 1         2         0     10  
## 2         1         0      9  
## 3         2         0      2  
## 4         2         0      2  
## 5         2         0      2  
## 6         2         0      2  
## # ... with 119,384 more rows
```

ascending / descending order

```
hotels %>%  
  select(adults, children, babies)  
  arrange(babies)
```

```
## # A tibble: 119,390 × 3  
##   adults children babies  
##   <dbl>     <dbl>  <dbl>  
## 1         2         0      0  
## 2         2         0      0  
## 3         1         0      0  
## 4         1         0      0  
## 5         2         0      0  
## 6         2         0      0  
## # ... with 119,384 more rows
```

```
hotels %>%  
  select(adults, children, babies)  
  arrange(-babies)
```

```
## # A tibble: 119,390 × 3  
##   adults children babies  
##   <dbl>     <dbl>  <dbl>  
## 1         2         0     10  
## 2         1         0      9  
## 3         2         0      2  
## 4         2         0      2  
## 5         2         0      2  
## 6         2         0      2  
## # ... with 119,384 more rows
```

filter

How could we remove the resort hotel? (hotel being "Resort Hotel")

```
hotels %>%  
  filter(____)
```

filter

How could we remove the resort hotel? (hotel being "Resort Hotel")

```
hotels %>%  
  filter(hotel != "Resort Hotel")
```

```
## # A tibble: 79,330 × 32  
##   hotel      is_canceled lead_time arrival_date_ye... arrival_date_mo.  
##   <chr>          <dbl>      <dbl>          <dbl> <chr>  
## 1 City Hotel      0         6          2015 July  
## 2 City Hotel      1        88          2015 July  
## 3 City Hotel      1        65          2015 July  
## 4 City Hotel      1        92          2015 July  
## 5 City Hotel      1       100          2015 July  
## 6 City Hotel      1        79          2015 July  
## # ... with 79,324 more rows, and 27 more variables:  
## #   arrival_date_week_number <dbl>,  
## #   arrival_date_day_of_month <dbl>,  
## #   stays_in_weekend_nights <dbl>, stays_in_week_nights <dbl>,  
## #   adults <dbl>, children <dbl>, babies <dbl>, meal <chr>,  
## #   country <chr>, market_segment <chr>,
```

`filter` keeps rows matching
conditions

Multiple conditions

Any bookings for only kids (no adults, but children or babies)?

Multiple conditions

Any bookings for only kids (no adults, but children or babies)?

```
hotels %>%  
  filter(  
    adults == 0,  
    children >= 1 | babies >= 1      # | means or  
  ) %>%  
  select(adults, babies, children)
```

```
## # A tibble: 223 × 3  
##   adults babies children  
##   <dbl>   <dbl>   <dbl>  
## 1      0      0      3  
## 2      0      0      2  
## 3      0      0      2  
## 4      0      0      2  
## 5      0      0      2  
## 6      0      0      3  
## # ... with 217 more rows
```

Logical operators in R

operator	definition	operator	definition
<	less than	<code>x y</code>	x OR y
<=	less than or equal to	<code>is.na(x)</code>	test if x is NA
>	greater than	<code>!is.na(x)</code>	test if x is not NA
>=	greater than or equal to	<code>x %in% y</code>	test if x is in y
==	exactly equal to	<code>!(x %in% y)</code>	test if x is not in y
!=	not equal to	<code>!x</code>	not x
<code>x & y</code>	x AND y		

Which market segments book most in advance?

market_segment	lead_time
Groups	186.973096
Offline TA/TO	135.004459
Online TA	82.998725
Direct	49.859115
Corporate	22.125590
Complementary	13.286676
Aviation	4.443038
Undefined	1.500000

How would we compute this?

Which market segments book most in advance?

market_segment	lead_time
Groups	186.973096
Offline TA/TO	135.004459
Online TA	82.998725
Direct	49.859115
Corporate	22.125590
Complementary	13.286676
Aviation	4.443038
Undefined	1.500000

```
hotels %>%  
  group_by(market_segment) %>%  
  summarize(  
    lead_time = mean(lead_time))
```

```
## # A tibble: 8 × 2  
##   market_segment lead_time  
##   <chr>          <dbl>  
## 1 Aviation        4.44  
## 2 Complementary   13.3  
## 3 Corporate       22.1  
## 4 Direct         49.9  
## 5 Groups        187.  
## 6 Offline TA/TO  135.  
## # ... with 2 more rows
```

Which market segments book most in advance?

market_segment	lead_time
Groups	186.973096
Offline TA/TO	135.004459
Online TA	82.998725
Direct	49.859115
Corporate	22.125590
Complementary	13.286676
Aviation	4.443038
Undefined	1.500000

```
hotels %>%  
  group_by(market_segment) %>%  
  summarize(  
    lead_time = mean(lead_time)) %>%  
  arrange(desc(lead_time))
```

```
## # A tibble: 8 × 2  
##   market_segment lead_time  
##   <chr>          <dbl>  
## 1 Groups          187.  
## 2 Offline TA/TO    135.  
## 3 Online TA        83.0  
## 4 Direct           49.9  
## 5 Corporate        22.1  
## 6 Complementary    13.3  
## # ... with 2 more rows
```

How many total nights for each booking?

```
hotels %>%  
  select(hotel, stays_in_week_nights, stays_in_weekend_nights)
```

```
## # A tibble: 119,390 × 3  
##   hotel          stays_in_week_nights stays_in_weekend_nights  
##   <chr>                <dbl>                <dbl>  
## 1 Resort Hotel          0                  0  
## 2 Resort Hotel          0                  0  
## 3 Resort Hotel          1                  0  
## 4 Resort Hotel          1                  0  
## 5 Resort Hotel          2                  0  
## 6 Resort Hotel          2                  0  
## # ... with 119,384 more rows
```

```
hotels %>%  
  mutate(  
    num_nights = ____)
```

How many total nights for each booking?

```
hotels %>%  
  mutate(  
    num_nights = stays_in_week_nights + stays_in_weekend_nights)  
  select(hotel, num_nights)
```

```
## # A tibble: 119,390 × 2  
##   hotel          num_nights  
##   <chr>          <dbl>  
## 1 Resort Hotel      0  
## 2 Resort Hotel      0  
## 3 Resort Hotel      1  
## 4 Resort Hotel      1  
## 5 Resort Hotel      2  
## 6 Resort Hotel      2  
## # ... with 119,384 more rows
```

How long did each market segment stay?

How long did each market segment stay?

```
hotels %>%  
  mutate(  
    num_nights = stays_in_week_nights + stays_in_weekend_nights)  
  group_by(market_segment) %>%  
  summarize(num_nights = mean(num_nights)) %>%  
  arrange(desc(num_nights))
```

```
## # A tibble: 8 × 2  
##   market_segment num_nights  
##   <chr>          <dbl>  
## 1 Offline TA/T0      3.90  
## 2 Aviation          3.61  
## 3 Online TA         3.57  
## 4 Direct            3.21  
## 5 Groups            2.99  
## 6 Corporate         2.09  
## # ... with 2 more rows
```

How many in each market segment was a repeating guest?

```
hotels %>% distinct(is_repeated_guest)
```

```
## # A tibble: 2 × 1
##   is_repeated_guest
##               <dbl>
## 1                 0
## 2                 1
```

How many in each market segment was a repeating guest?

```
hotels %>% distinct(is_repeated_guest)
```

```
## # A tibble: 2 × 1
##   is_repeated_guest
##   <dbl>
## 1             0
## 2             1
```

```
hotels %>%
  group_by(market_segment) %>%
  summarize(frac_repeat = mean(is_repeated_guest)) %>%
  ggplot(aes(y = fct_reorder(market_segment, frac_repeat), x = frac_repeat))
  geom_col()
```

